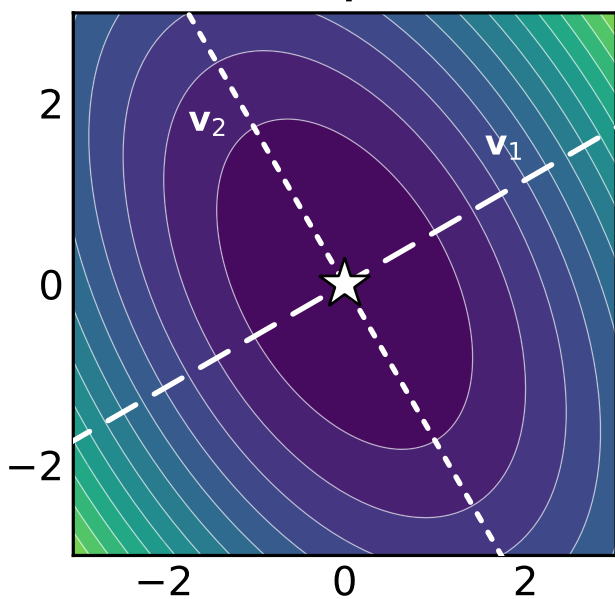
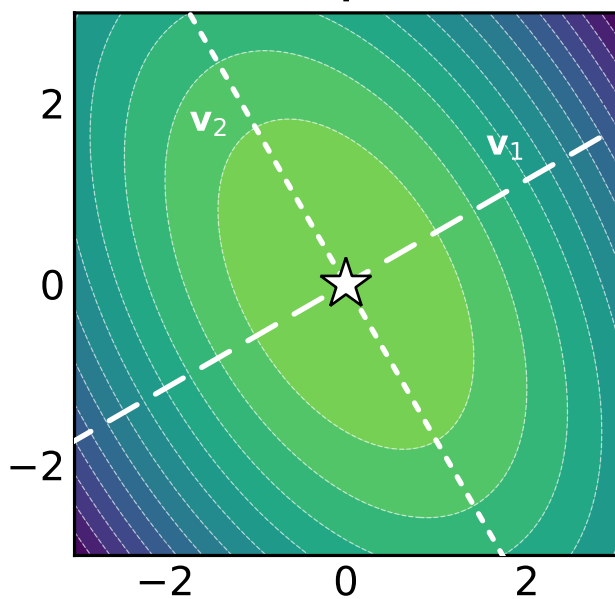


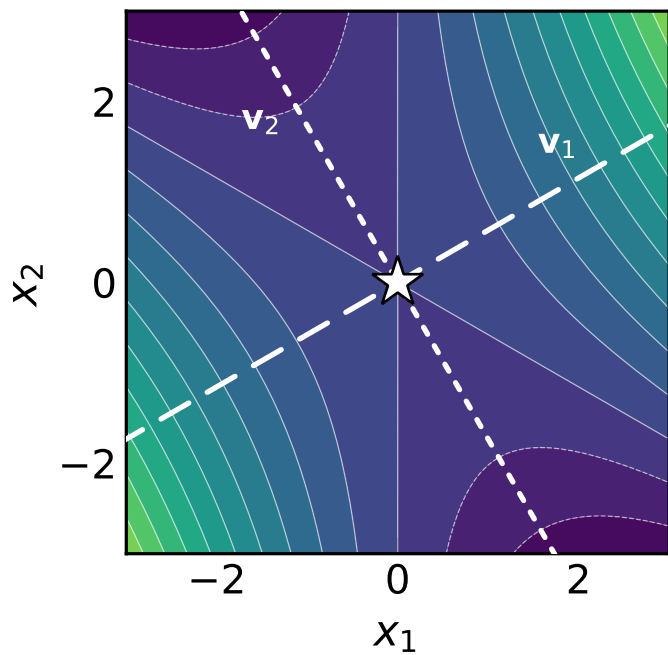
$\lambda = (3, 1)$ — **positive definite**
 $\mathbf{x}^*$ is the unique minimum



$\lambda = (-3, -1)$ — **negative definite**
 $\mathbf{x}^*$ is the unique maximum



$\lambda = (3, -1)$ — **indefinite**
 $\mathbf{x}^*$ is a saddle point



$\lambda = (2, 0)$ — **positive semidefinite**
 $\mathbf{x}^*$ is one of a line of minima

